

(ii)	$U = \rho V g$ $= (1.3) \left(\frac{4}{3} \pi \left(\frac{1.5}{2} \right)^3 \right) (9.81) = 22.54 N$
	$U = T + mg$ $T = U - mg$ $= 22.54 - (0.3)(9.81) = 19.6 N$
(b)(ii)	Net force upwards as upthrust is more than weight
	As altitude increases, air pressure/density decreases, causing upthrust to decrease
	Upthrust will decrease till its magnitude is equal to weight.

4(a)	e.m.f. is the amount of non-electrical energy converted into electrical energy per unit charge passing through the terminals of the cell (source).
	p.d. is the amount of electrical energy converted to other forms of energy per unit charge passing from one point to the other.
(b)(i)	Draw best fit line. $\text{Emf} = 9.0 \text{ V}$
(b)(ii)	Read off two points along best fit line. (0, 9.0) and (1.800, 5.4)
	$V = E - Ir$ $5.4 = 9.0 - 1.8(r)$